INCH-POUND
MIL-C-38404E
16 October 1992
SUPERSEDING
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6 November 1985

MILITARY SPECIFICATION

COUPLINGS, HOSE, REATTACHABLE SCREW-ON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers screw-on type reattachable couplings for hoses conforming to MIL-H-6615, or MIL-H-26521, as specified (see 6.1).

1.2 Classification. The couplings shall be of the following classes and types, as specified (see 6.1):

Class 1	Fuel service
Type I	External NPT thread
Type II	Internal NPSH thread with swivel
Class 2	Demineralized water service
Type I	External NPT thread
Type II	Internal NPSH thread with swivel

1.3 Part numbering. Definitive part numbers shall be accordance with 6.4.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TIRDM, Kelly AFB, TX 78241-5606, by using the self-addressed Standardization document Proposal DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730
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SPECIFICATIONS

FEDERAL

ZZ-H-466	Hose Assembly, Metal (Rubber Covered), Hydrocarbon Conveying.

MILITARY

MIL-H-775	Hose, Hose Assemblies, Rubber, Plastic, Fabric or Metal (including Tubing); and Associated Hardware: Packaging of
MIL-H-6615	Hose Assemblies, Rubber, Fuel and Water, with Reattachable Couplings, Low Temperature.
MIL-R-6855	Rubber, Synthetic, Sheets, Strips Molded or Extruded Shapes, General Specification for,
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-H-26521	Hose and Hose Assembly, Nonmetallic, Fuel, Collapsible, Low Temperature with Reattachable Couplings.

STANDARDS

FEDERAL

FED-STD-H28	Screw-Tread Standards for Federal Services
FED-STD-595	Colors Used in Government Procurement

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.1	Alloy and Temper Designation System for Wrought Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

ASTM B-283	Standard Specification, Copper and Copper-Alloy Die Forgings (Hot Pressed)
ASTM B-633	Electrodeposits zinc
ASTM D-3951	Commercial Packaging of Supplies and Equipment

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see sections 4 and 6).

3.2 Components. The coupling shall consist of the following major components:

Item name

Body (see 3.7.1)

Grip (see 3.7.2)

Sleeve (see 3.7.3)

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.4 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service and shall be entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished coupling.

* 3.4.1 Material for class 1 couplings. The material for class 1 couplings shall be as follows:

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Body - High quality high-grade forging brass in accordance with ASTM B-283 copper alloy UNS No. C37700 containing not less than 58 percent copper, or equivalent.

Sleeve - High quality high-grade brass castings or forgings containing not less than 58 percent copper.

Grip - Electrodeposited zinc in accordance with ASTM B-633, type II, SC1.

3.4.2 Material for class 2 couplings. The material for class 2 couplings shall be as follows:

Body and sleeve - 300 series stainless steel, type B214 or 356 aluminum casting alloy, or aluminum alloys other than 2000, 4000, and 7000 series in accordance with ANSI H35.1. Aluminum alloys shall be anodized in accordance with MIL-A-8625. The color shall be olive drab, color number 24087 of FED-STD-595.

Grip - Electrodeposited zinc in accordance with ASTM B-633, type II, SC1

3.5 Design and construction. The couplings shall be designed and constructed for use with hoses conforming to MIL-H-6615 or MIL-H-26521. They shall be reattachable, shall incorporate the feature specified herein and as shown on Figure 1, and shall require no cements, sealants, or adhesives in their installation, other than to seal the cut ends of the hose.

3.5.1 Thread gages. All threads shall be in accordance with Handbook H28. If the male tailpiece threads and female sleeve threads are in accordance with the Handbook H28 special threads (NS or UNS), thread gages shall be provided by the manufacturer. Data specifying the accuracy of the thread gages shall be made available to the procuring activity. (See 3.7.1.1, 3.7.1.2, 3.7.1.3, and 3.7.3).

3.6 Performance.

3.6.1 Environmental conditions. The couplings shall exhibit no corrosion after having been exposed to the following environmental conditions:

a. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost.

b. Salt atmosphere as encountered near bodies of salt water and in transportation on shipboard.

3.6.2 Hydrostatic pressure rating. The internal hydrostatic pressure rating shall be as specified in Table 1. The size of the coupling shall be as specified by the procuring activity (see 6.4).

3.6.3 Hose retention. The couplings shall be installed on the hose and shall not be detached from the hose when subjected to a repeated longitudinal load. After the repeated load test, couplings shall not blow off or leak when test pressure specified in Table I.

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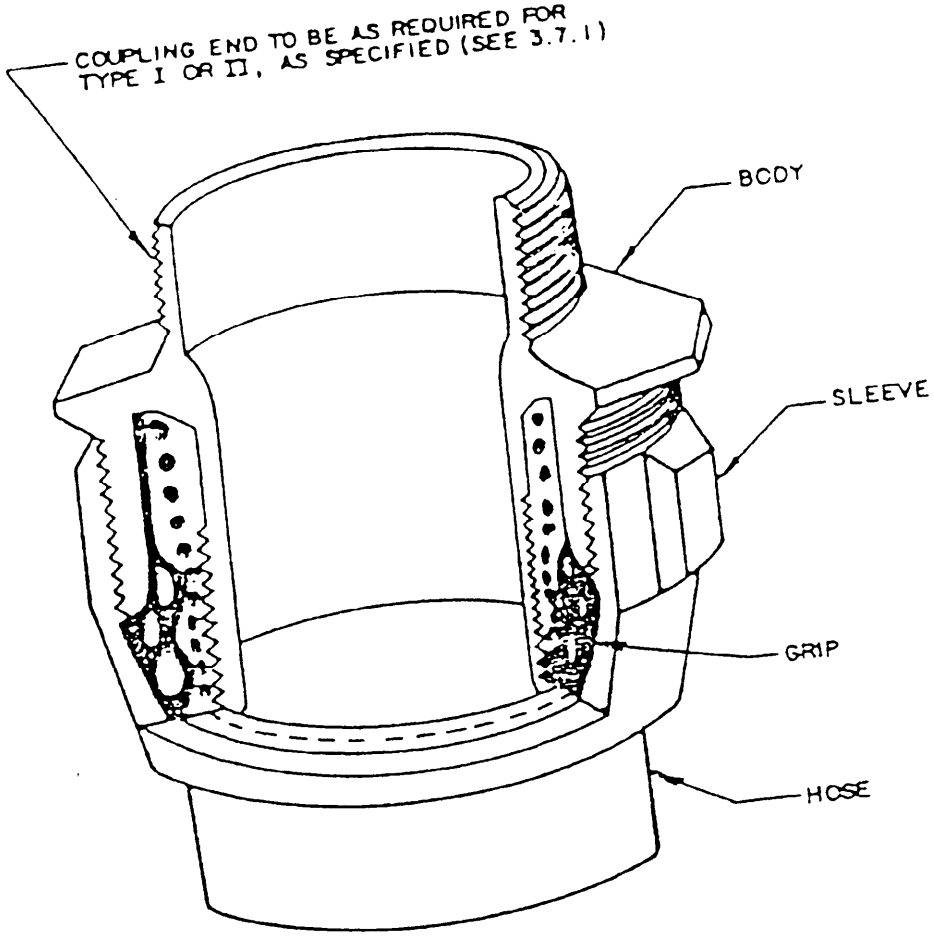


FIGURE 1. Section showing typical coupling assembly construction.

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3.2.2 Standard parts. With the exception specified in [illegible]

TABLE I. Pressure rating.

Size	Size Code	Working Pressure	Test pressure
1.25	A	400	600
1.50	B	400	600
2	C	380	570
2.50	D	360	540
3	E	320	480
4	F	280	420

3.7 Detail of components.

3.7.1 Body. The body of the coupling shall have a configuration as shown on Figure 2. The body shall have a coupling end for attachment to mating assemblies and a tailpiece end for attachment of the applicable hose. The central hexagon or octogen shall have a larger dimension across the flats than the largest diameter across corners of the coupling sleeve.

3.7.1.1 Type I coupling end. The type I coupling end shall have external American Standard taper pipe threads (NPT). The end surface of the coupling end shall be a smooth finish with rounded edges, square with the thread axis, to provide a good sealing surface when joined to a type II coupling.

3.7.1.2 Type II coupling end. The type II coupling end shall have internal American National hose coupling threads (NPSH) which shall swivel on the remainder of the body. The 2.5-, 3-, and 4 inch nominal size type II couplings shall conform to the 8 threads per inch sizes of ZZ-H-466. The body shall provide a smooth sealing surface for a gasket. Each body shall be furnished with a gasket of synthetic rubber conforming to MIL-R-6855, class 1. The swivel end shall turn freely by hand, without the use of tools or mechanical assistance, until slightly before a leaktight seal is effected against the gasket.

* 3.7.1.3 Tailpiece end. The tailpiece end shall consist of two concentric walls spaced to accept hose in accordance with MIL-H-6615 or MIL-H-26521, of the appropriate size. The outer wall of the tailpiece end shall have external threads to engage the sleeve and have a thickness of at least 75 percent of the appropriate size hose. The external threads shall be straight and have 14 threads per inch. The taper shall extend through at least 75 percent of the wall thickness at an angle between 45 and 50 degrees. The A and B sizes shall have a minimum of 6 complete threads, C and D sizes shall have a minimum of 13 complete threads, and E and F a minimum of 19 complete threads (see Table I for sizes). The inner wall of the tailpiece end shall have ridges to assure a tight grip on the hose, a rounded outer edge to prevent cutting of the hose during installation or upon sharp bending, and a thickness of at least 30 percent of the appropriate size hose. The ridges shall be cut to a depth between 30 and 45 percent of the inner wall thickness with

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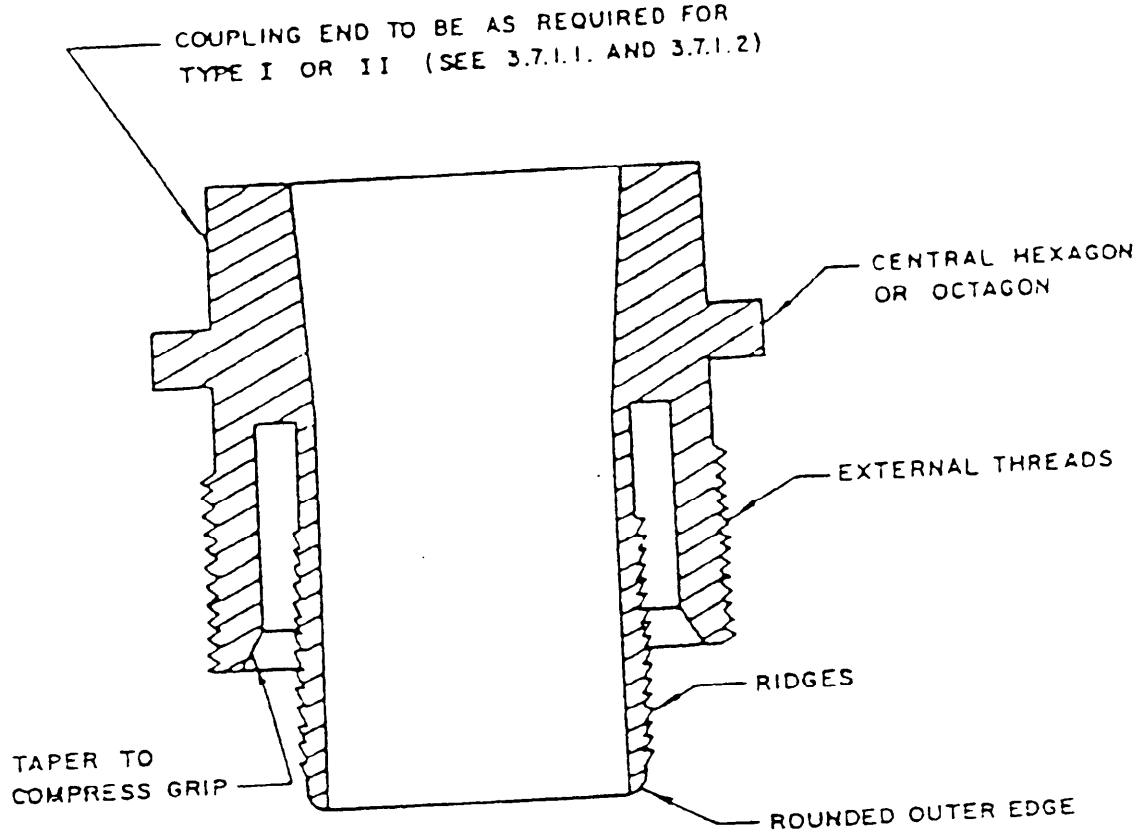


FIGURE 2. Typical cross-section of coupling body.

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sharp edges having a diameter equal to that of the inner wall outer diameter. Truncated edges are not acceptable. The first ridge shall begin no farther out than where the taper begins on the inside of the outer wall. The last ridge shall end at least four ridges out from the open end of the outer wall. The inner wall shall be at least 175 percent as long as the outer wall. (When thickness involves threads, the measurement shall be from the thread tip to the wall).

3.7.2 Grip. The grip shall be of material as specified in 3.4.1 and 3.4.2. The wire shall be left-hand wound into a coil of approximately three loops, and may be round or flat with rounded edges. The end of the grip shall be ground to prevent damaging the hose or injuring using personnel during installation or removal of the coupling.

* 3.7.3 Sleeve. The sleeve shall have a configuration incorporating the features shown on Figure 3 and as specified herein. Four flat surfaces (spaced 90 degrees apart) shall be provided for attaching a wrench. The interior of the sleeve shall have internal threads for connection with those provided on the body, shall be tapered to compress the grip, and shall have the edge rounded to prevent cutting of the hose upon sharp bending. The grip compression taper angle shall be between 50 and 65 degrees. The exterior shall taper from where the interior threads stop to the rounded edge end of the sleeve in a smooth continuous taper. When fully threaded to the body, the sleeve shall extend beyond the body tailpiece.

3.8 Interchangeability. All functionally and dimensionally interchangeable part shall have the same manufacturer's part number. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part numbers.

* 3.9 Marking. The body specified in 3.7.1 and sleeve specified in 3.7.3 shall be Marked (impression stamped) with the applicable hose size, manufacturer's code, and MIL-Spec (MIL-C-38404) in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The examination and testing of the couplings shall be classified as follows:

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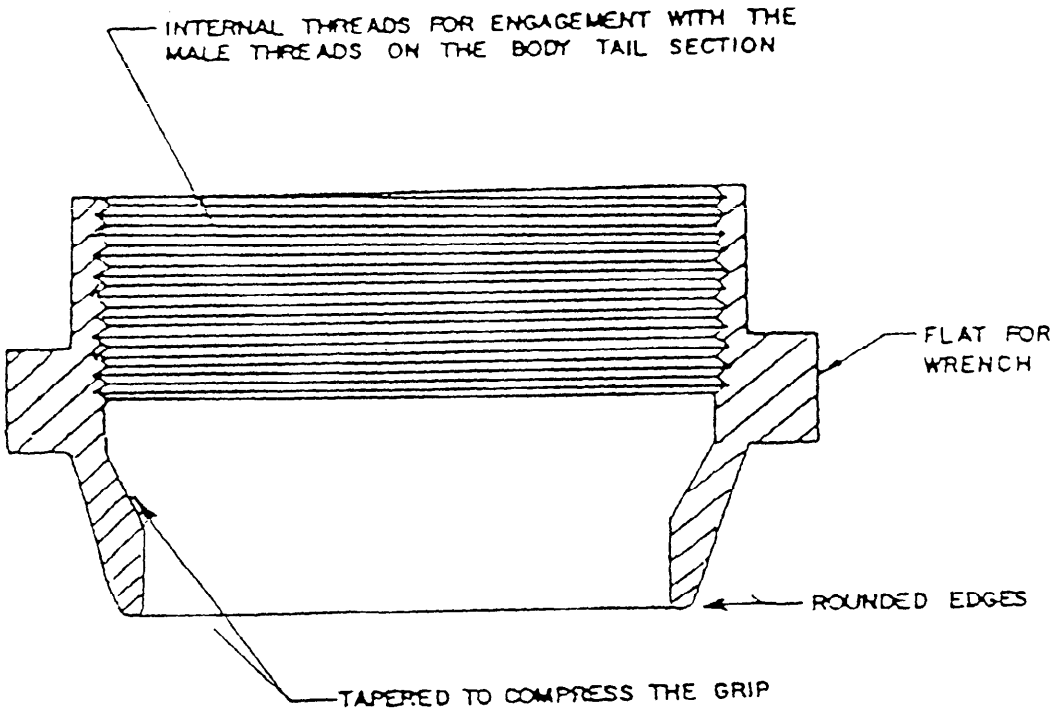


FIGURE 3. Cross-section of coupling sleeve.

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a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection

* 4.3.1 Test sample. One coupling for each size of each class and type being procured for each applicable hose specification shall be subjected to the tests specified in 4.3.3.

* 4.3.2 Test report. Upon completion of the first article tests, three completed copies of a test report in accordance with MIL-STD-831 and one coupling of each tested size and type shall be supplied to the procuring activity.

4.3.3 First article inspection. The first article inspection shall consist of all the tests described under 4.5.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the individual tests.

4.4.1 Individual tests. Each coupling shall be subjected to the examination specified in 4.5.1. If the body is cast, each coupling shall be subjected to the test specified in 4.5.2.

4.5 Inspection methods.

4.5.1 Examination of product. The coupling shall be inspected to determine compliance with the requirements specified herein. The following shall be applicable to the first article item only and shall be included in the test report:

a. Weight of component.
b. Dimensions of components.
c. Certification that the materials used complied with the requirements specified herein.

4.5.2 Pressure test. The coupling body shall be subjected to the applicable internal hydrostatic test pressure specified in Table I for not less than 1 minute. Evidence of leakage shall be cause for rejection.

4.5.3 Swivel test. When applicable, the internally threaded and connection of the body shall demonstrate the ability to swivel by hand relative to the tailpiece which is attached to the hose. After this demonstration, the coupling assembly shall be subjected to the test specified in 4.5.4.

4.5.4 Hose retention test. The coupling assembly shall be attached to a six foot long section of hose conforming to MIL-H-6615 or MIL-H-26521. The hose assembly shall be hung vertically by any convenient means and be used to lift a dead weight of 450 pounds at least one inch off the floor and then lowered. This cycle shall be repeated 25 times. The assembly shall then be subjected to the test pressure specified in Table I for a minimum of one minute. Any evidence of leakage, pull or blow off of the coupling shall be cause for rejection.

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4.5.5 Environmental tests. The coupling shall be tested in accordance with 4.5.5.1 through 4.5.5.3.

4.5.5.1 Humidity. The coupling shall be subjected to humidity in accordance with procedure 1, method 507, of MIL-STD-810.

4.5.5.2 Salt spray. The coupling shall be subjected to salt spray in accordance with procedure I, method 509, of MIL-STD-810.

4.5.5.3 Verification with substantiating data that materials used are not adversely affected by humidity or corrosion may be provided in lieu of the humidity and salt spray testing.

4.6 Packaging, packing, and marking. Preparation for delivery shall be inspected for conformance to section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with applicable levels of MIL-H-775 or ASTM D-3951, as specified (see 6.1).

5.2 Marking. In addition to any special marking required by the contract or order (see 6.3), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The couplings covered by this specification are intended for use on hose conforming to MIL-H-6615 or MIL-H-26521 in ground fueling of aircraft with hydrocarbon fuels (use class 1) or demineralized water (use class

6.2 Electrodeposits zinc is not recommended for uses where prolong exposure to salt atmospheres is anticipated.

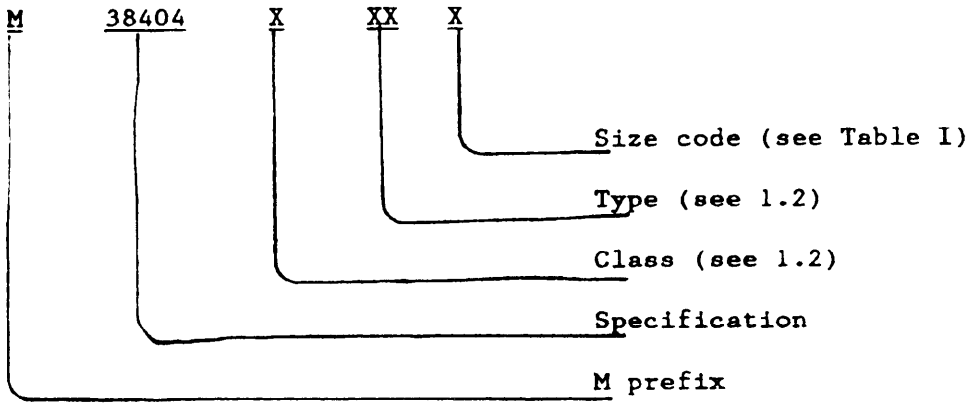
6.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class, type and size (see 1.2, 3.5, 3.6.2, 3.7.1, and figure 1).
c. Location and conditions for first article inspection (see 4.3).
d. Applicable levels of preservation, packaging, and packing required (see section 5.1).
e. Special shipment marking (see 5.2).

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6.4 Order of precedence. [illegible]

6.4 Part number. Definitive part number shall be in accordance with the following:



Example:

M 38404-2-II-C

Class 2, type II, 2 inch I.D.

6.5 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor's are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-38404E	92 OCT 16

3. DOCUMENT TITLE
COUPLINGS, HOSE, REATTACHABLE SCREW ON

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial 512-925-6314	(2) AUTOVON 945-6314
c. ADDRESS (Include Zip Code) SA-ALC/TIRDM KELLY AFB TX 78241-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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a. Operating pressure [illegible]

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